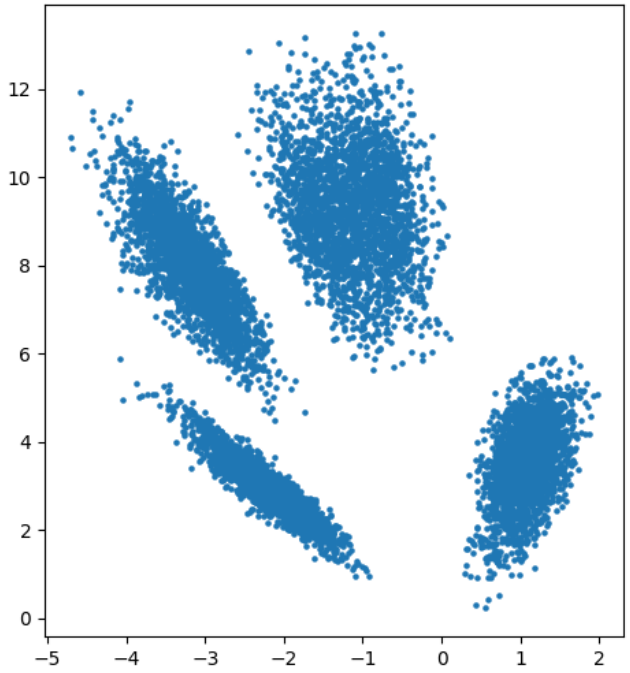
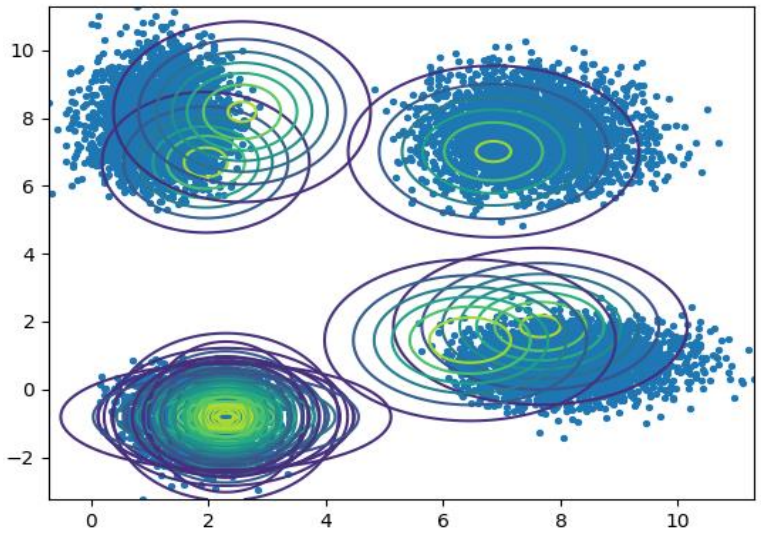
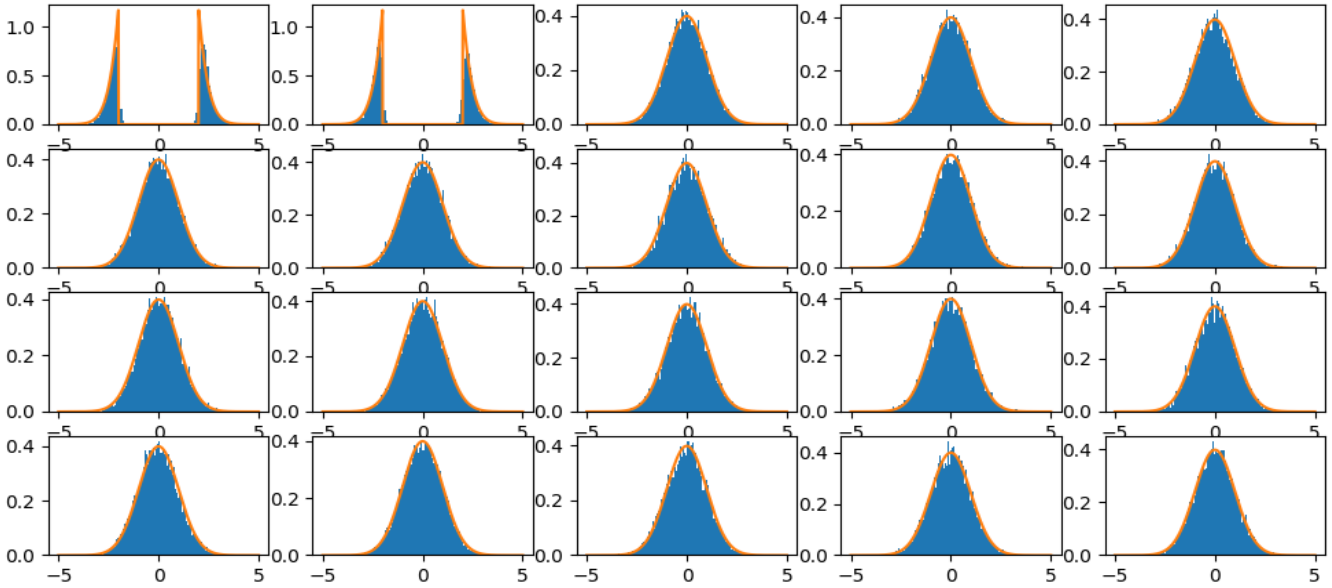
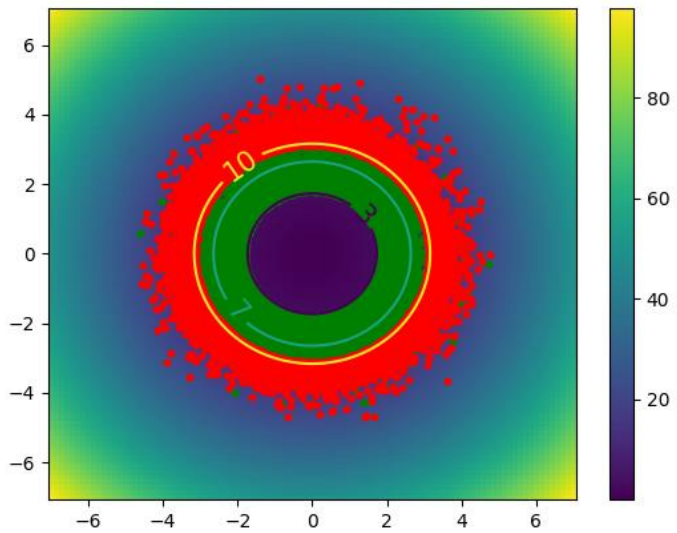
Entrainement VAE pour la distribution d'un vecteur gaussien tronqué sur les 2 premières composantes : 4 modes et 10 000 données

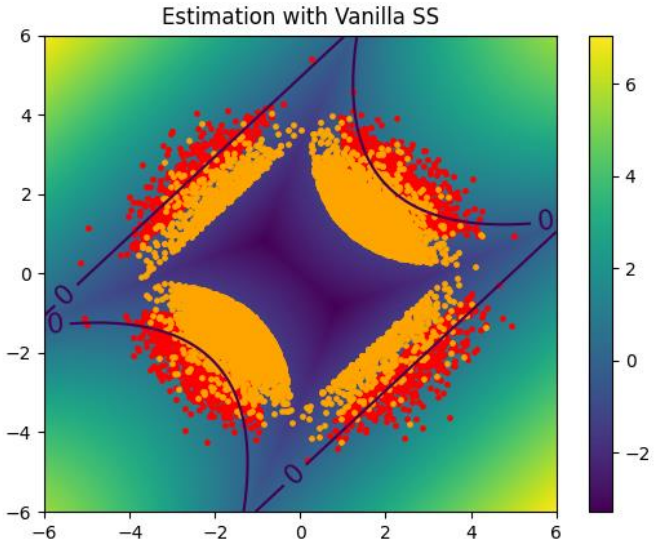
Melange gaussian de 15 composantes

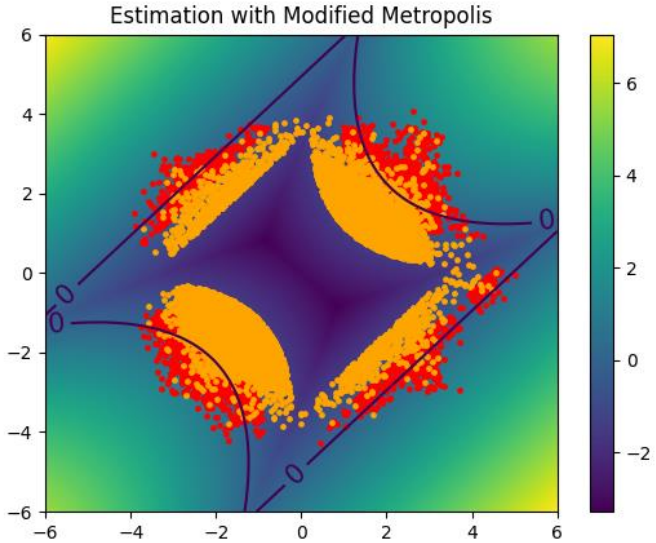


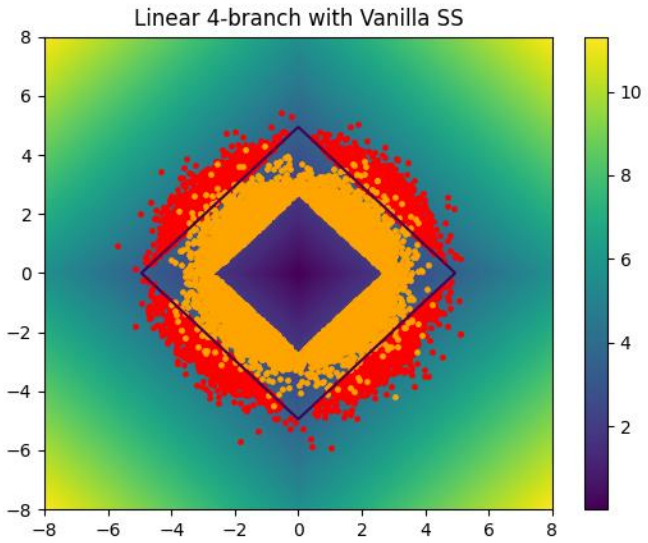


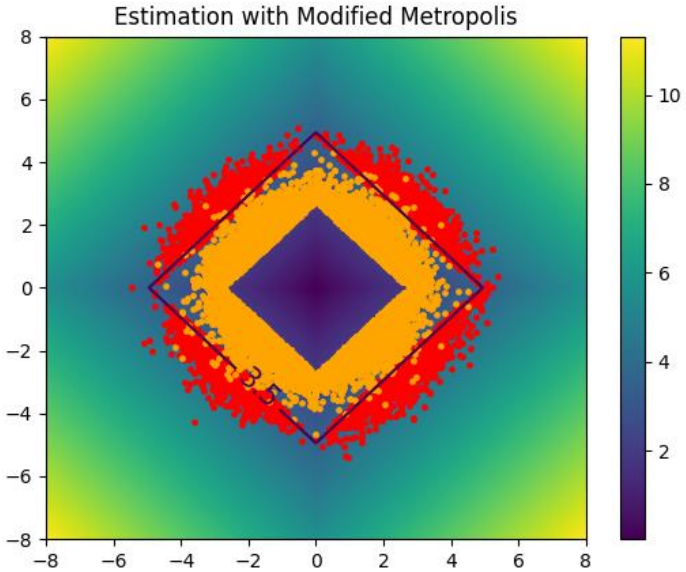


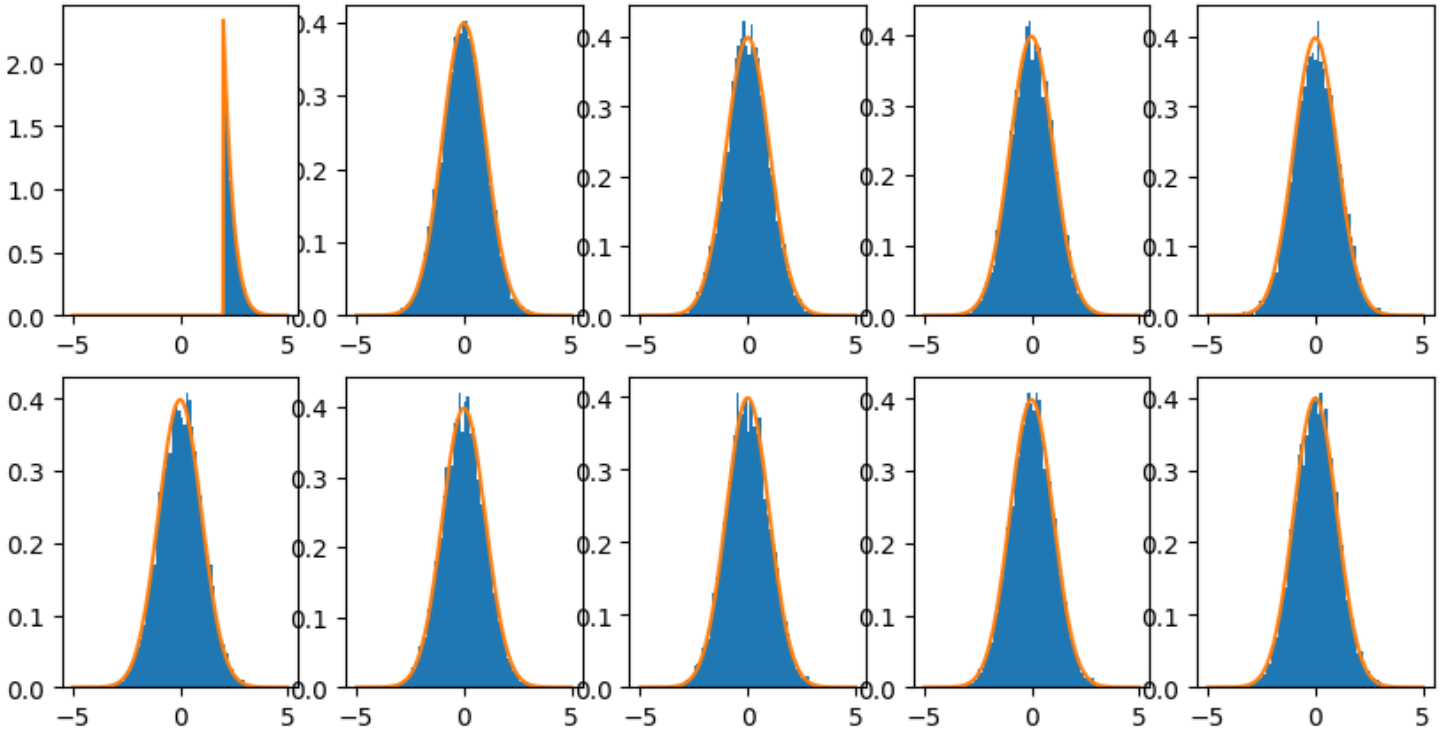

10
7
3


Estimation with Vanilla SS
0

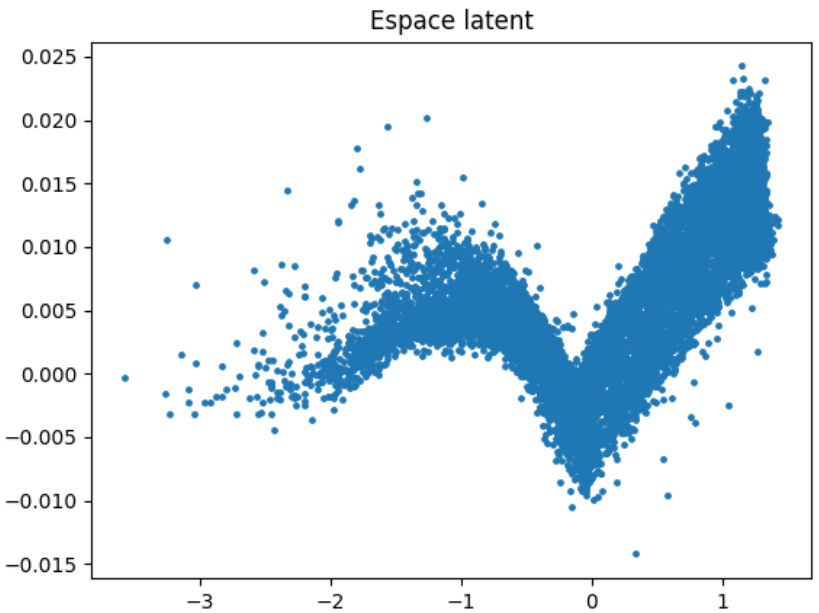

Estimation with Modified Metropolis
0


Linear 4-branch with Vanilla SS


Estimation with Modified Metropolis
3.5

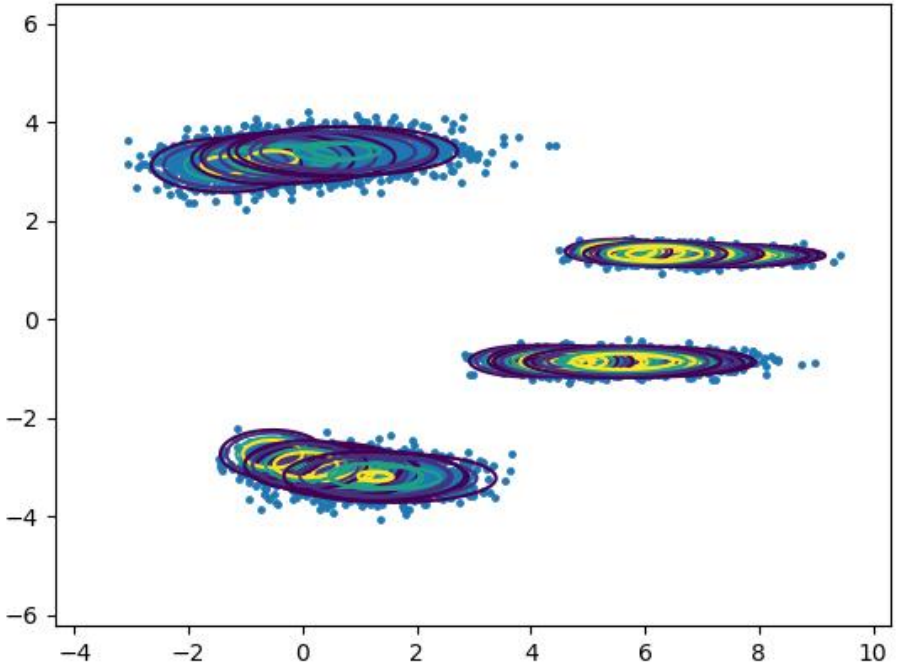


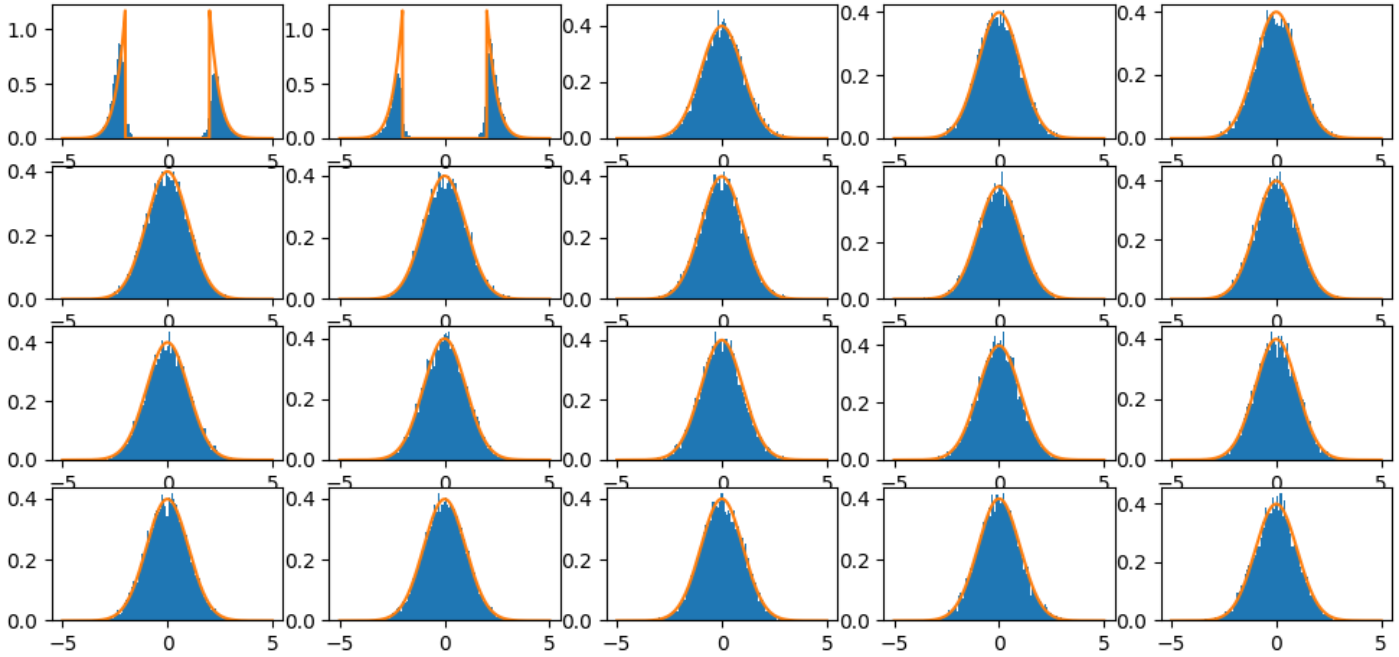
Cas de la troncature mais pour seulement une composante et un coté : Utilisation d'un vae avec un prior normal standard

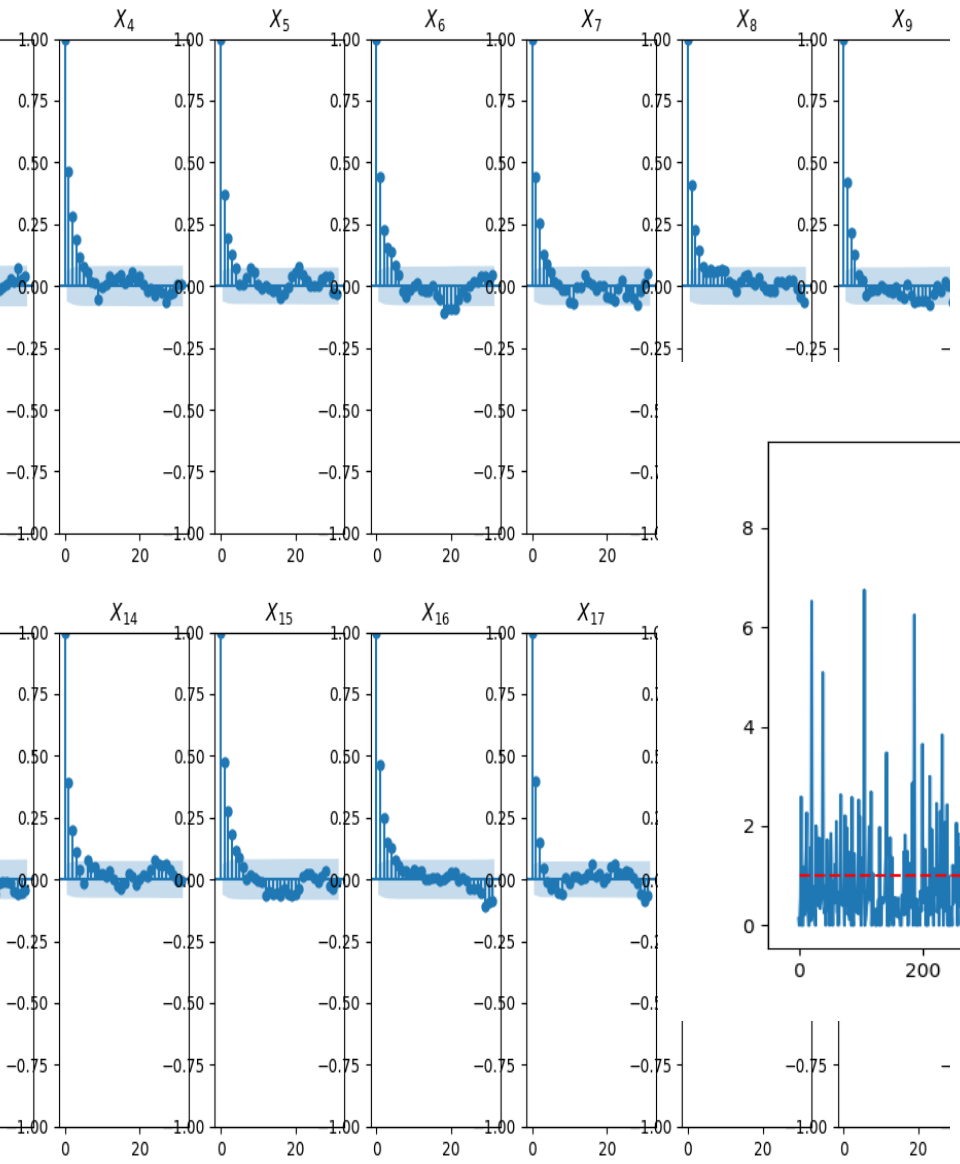


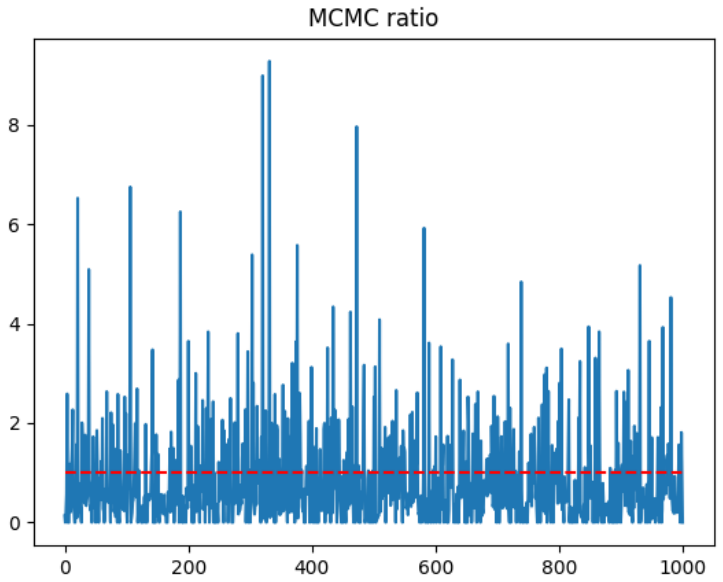
Vamprior

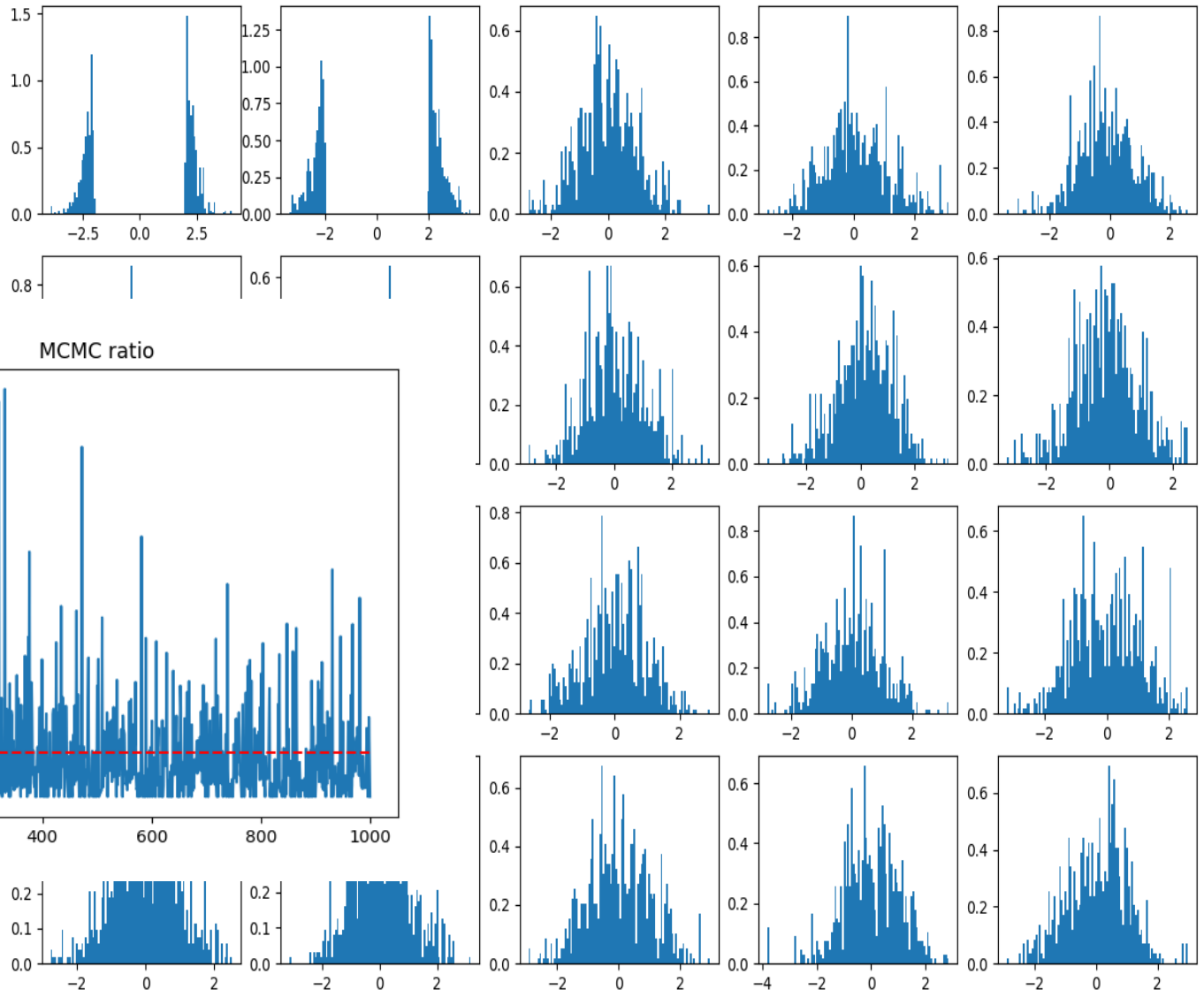
Taux acceptation : autour de 0,5

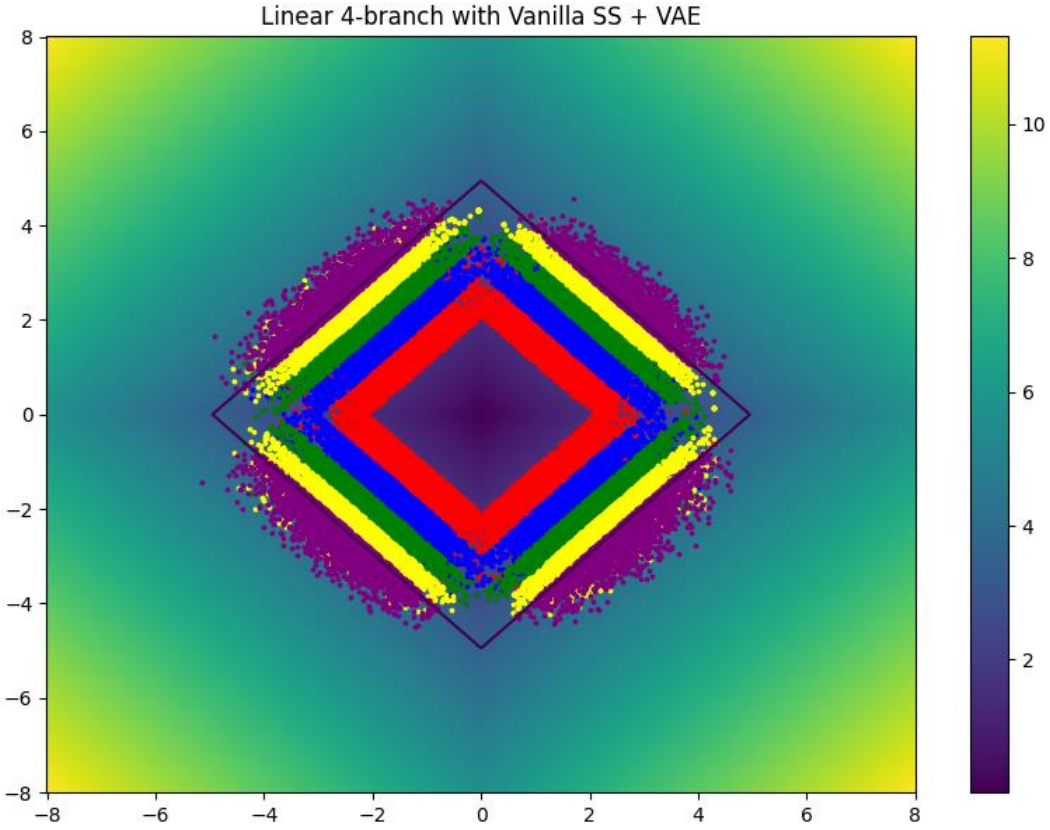


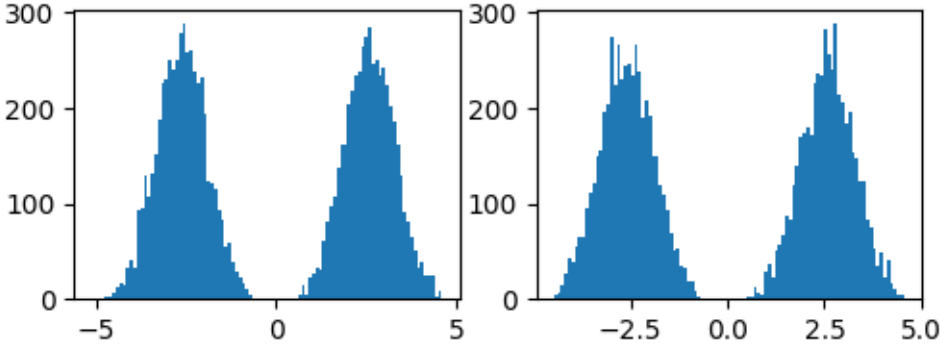



X_4
X_5
X_6
X_7
X_8
X_9
X_{14}
X_{15}
X_{16}
X_{17}


MCMC ratio

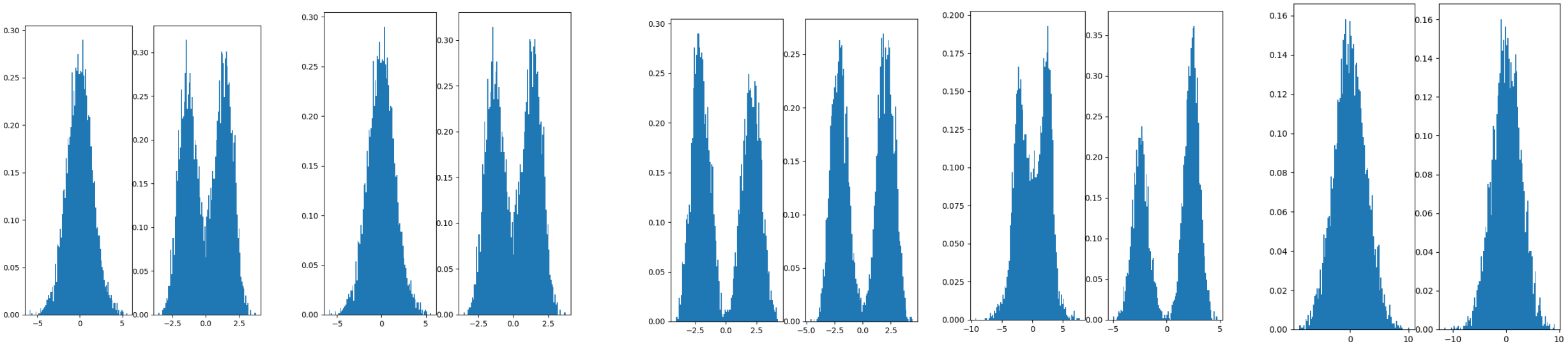




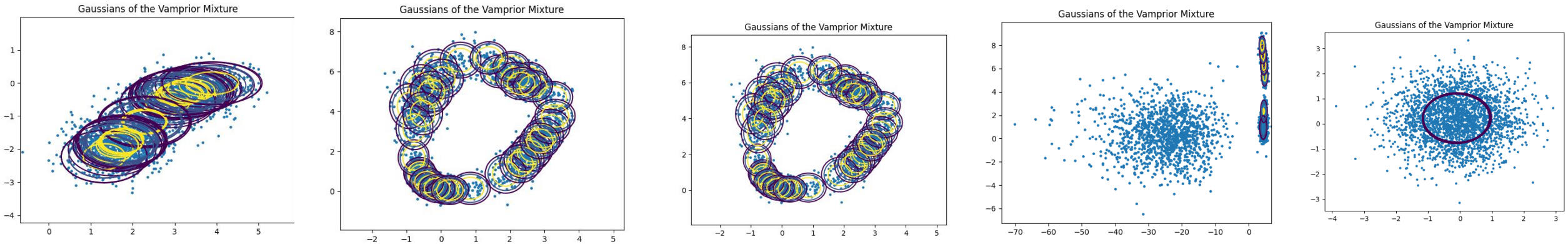


Exemple d’un bon apprentissage en dimension 2 avec un espace latent de dimension 2

Cas de la dimension 2 avec un espace latent de dimension 2



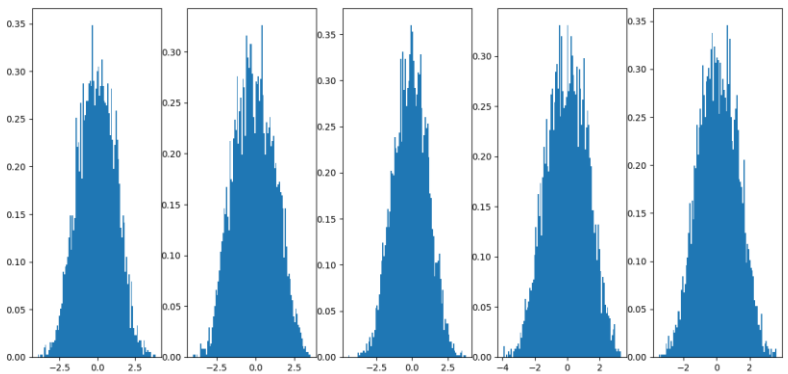
Augmentation du seuil de défaillance

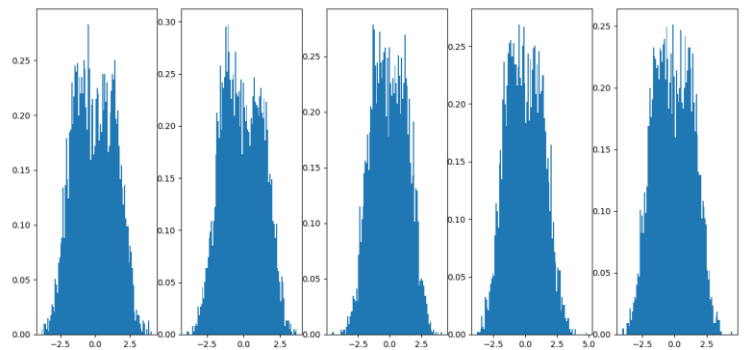


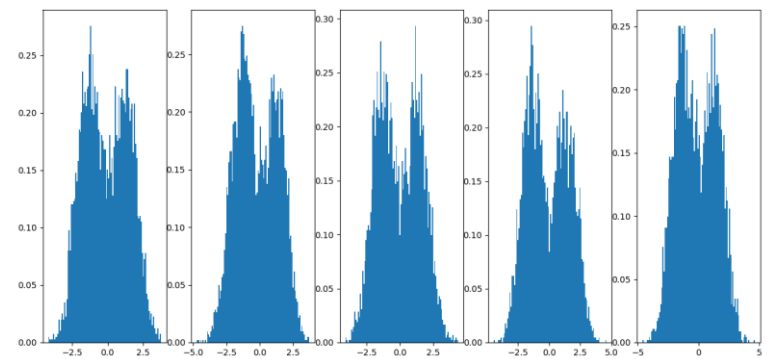
VAE Vamprior sur X dans R2 ; espace de latent de dimension 2, 35 pseudo-inputs, proba fixe p0 = 0,25 soit 2500 échantillons pour le VAE

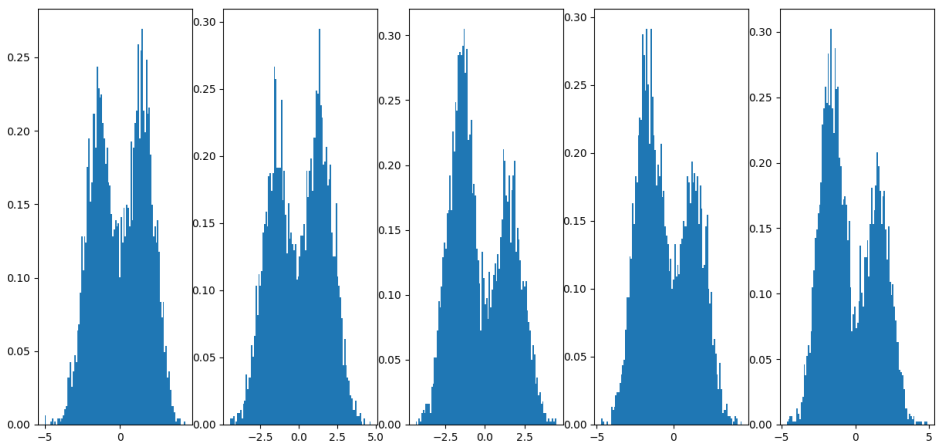
Fail de l'encodage pour les événements les + défaillants
Estimation Pf = 0.000925

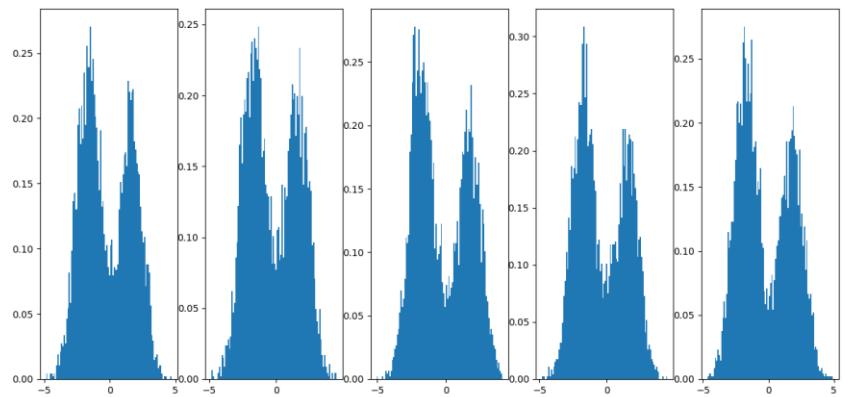
Cas de la dimension 5 avec un espace latent de dimension 3



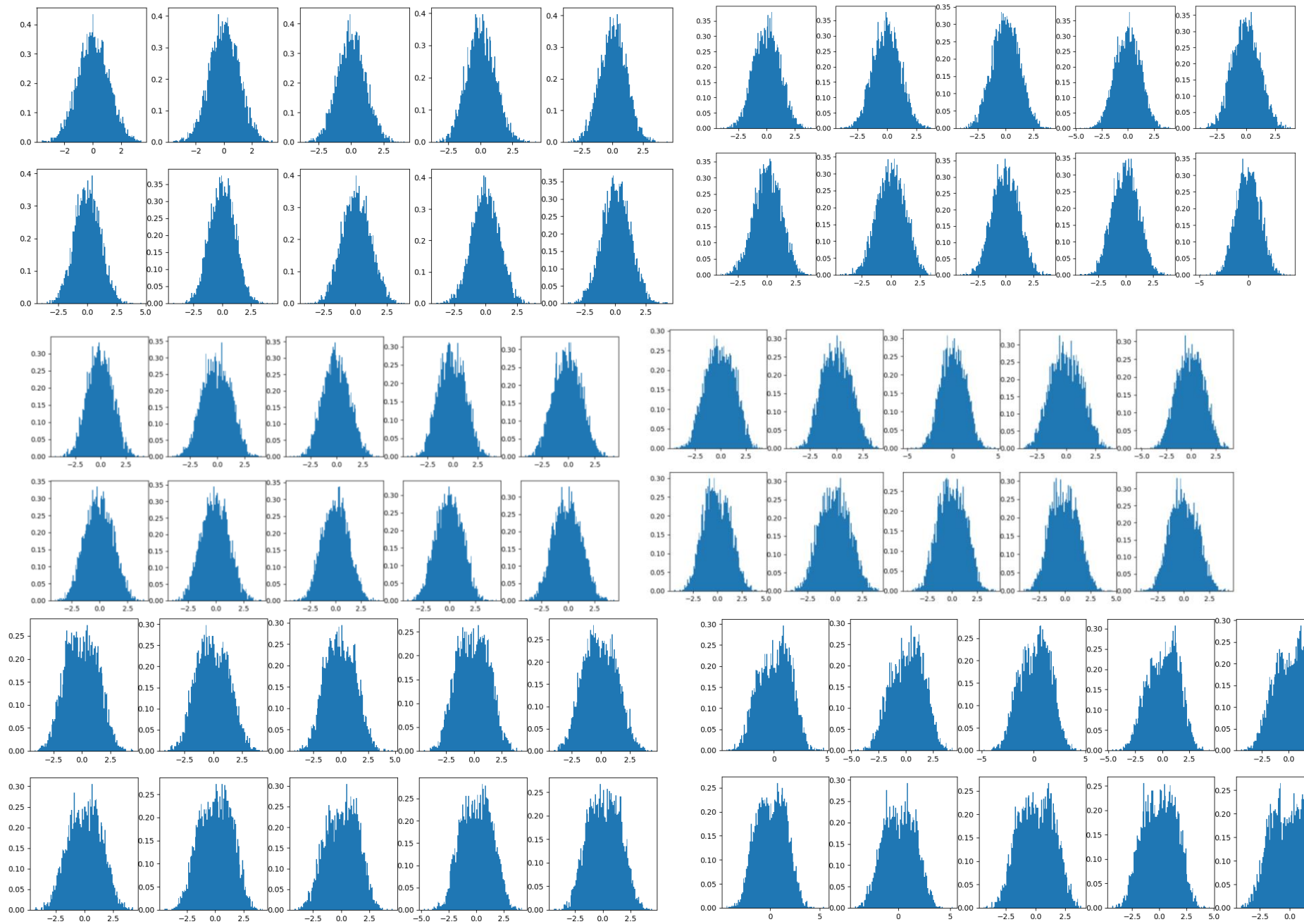








Cas de la dimension 10 avec un espace latent de dimension 3, 45 pseudo_inputs

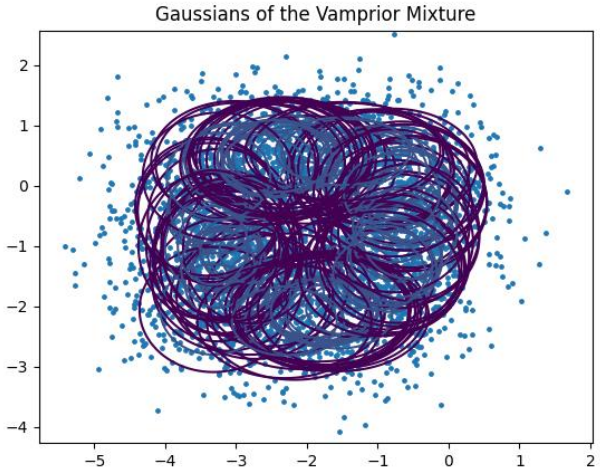


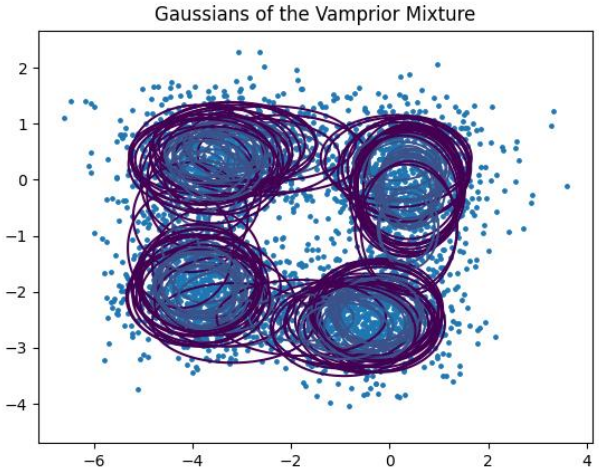
Failure : 0.000933

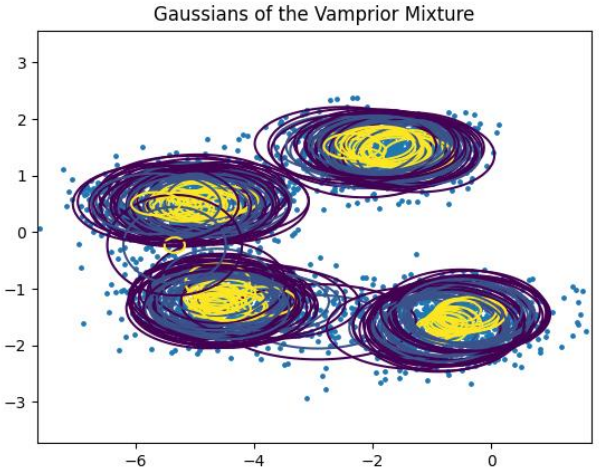
Quantile [1.29, 1.84, 2.28, 2.67, 3.00, 3.31, 3.60]

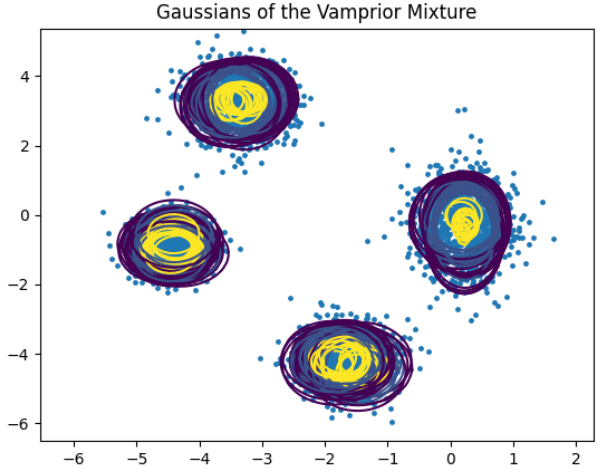
Taux d'acceptation [0.3498 0.267 0.23218333 0.20425 0.18686667 0.17426667]

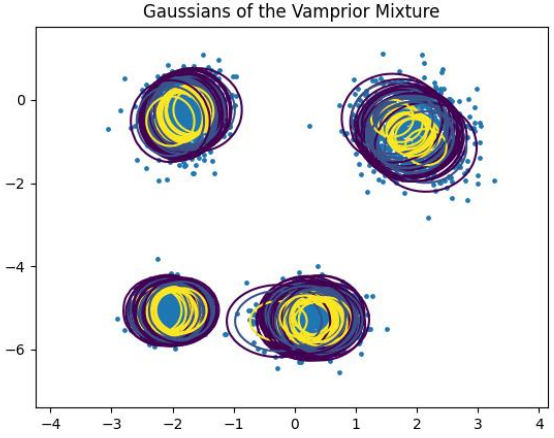
75 Pseudo-inputs + p0 = 0,35

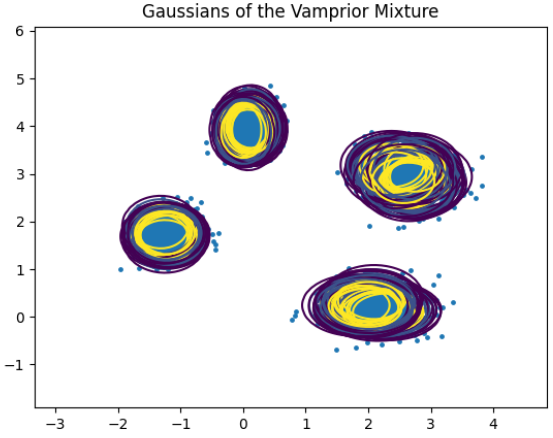

Gaussians of the Vamprior Mixture


Gaussians of the Vamprior Mixture


Gaussians of the Vamprior Mixture


Gaussians of the Vamprior Mixture


Gaussians of the Vamprior Mixture


Gaussians of the Vamprior Mixture

On utilise un pourcentage de 25% pour chaque étape du VAE

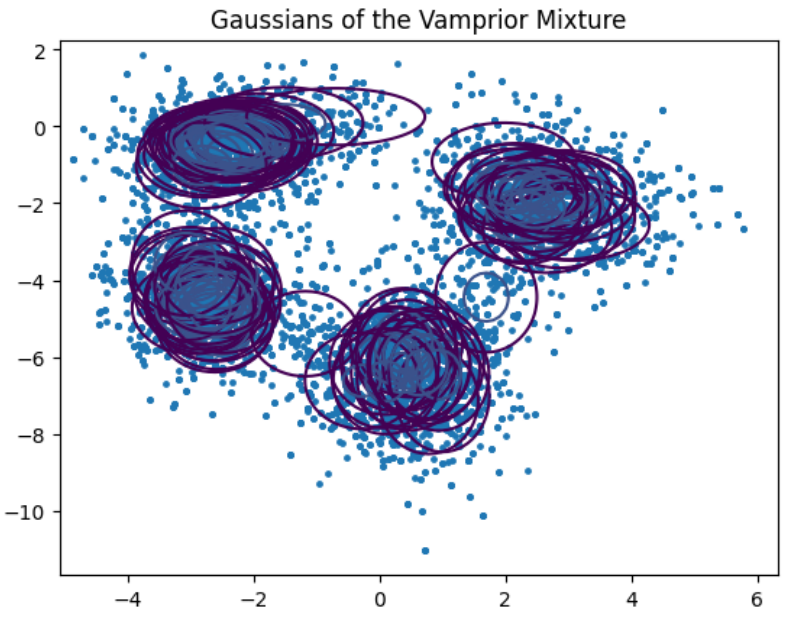
Utilisation de la même souche pour les données

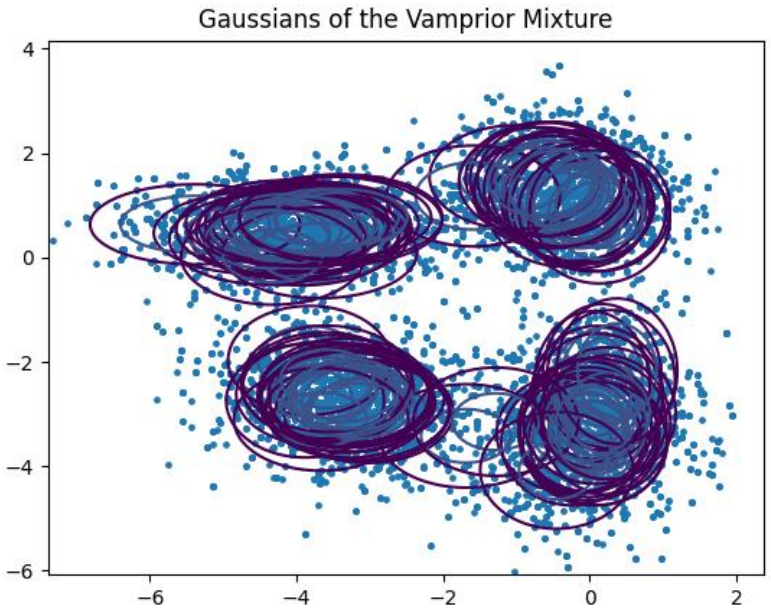
Pf = 0,00093

Dimension	MMA	CMC	Vanilla	SS + VAE	SS + VAE Param
D = 2	Pf = 0,0009	Pf = 0,0007	Pf = 0,0087	Pf = 0,00087	K = 35, Latent =2 Time = 2s/boucle
D= 10	Pf = 0,00093	Pf = 0,0014	Pf = 0,00111	Pf = 0,00101	Time = 13s/boucle
D = 20	Pf = 0,00086	Pf = 0,0008	Pf = 0,0009	Pf = 0,001	Time = 18s/boucle
D = 50	Pf = 0,00092	Pf = 0,0007	Pf = 0,00086	Pf = 0,00122	Time = 1min/boucle
D = 70	Pf = 0,00094	Pf = 0,0004	Pf = 0,00086	Pf = 0,00152	Time = 2min/boucle
D = 100	Pf = 0,00093	Pf = 0,0012	Pf = 0,00081	Pf = 0,00184	Time = 3min/boucle

La partie évaluation du mélange infinie prend bcp de temps Trouver une solution..

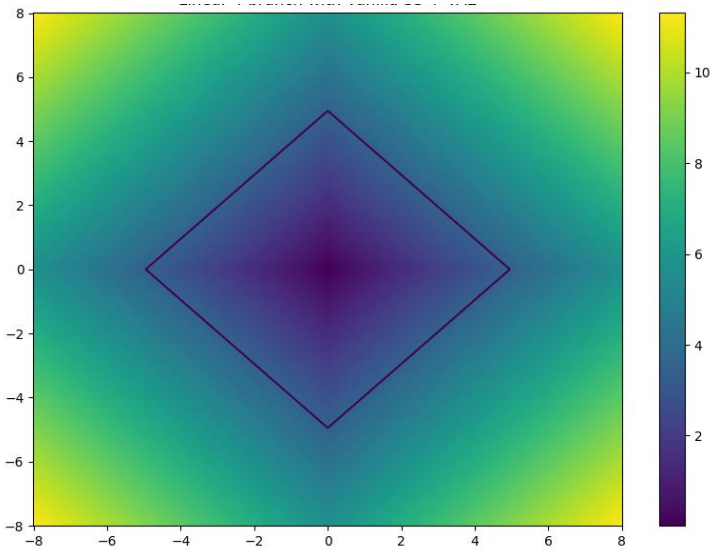
Comparaison VAE Julien



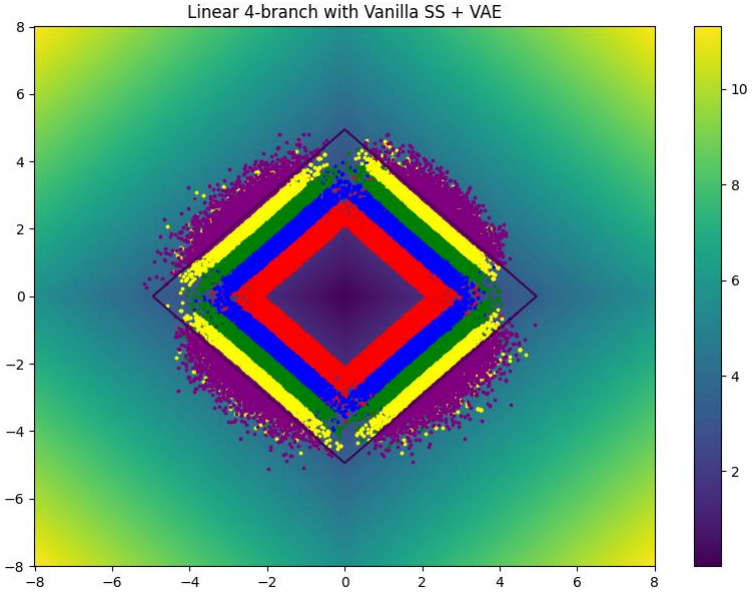


Test similarité VAE dans le cas du 4-Branches avec un X gaussian de dimension 50 et un échantillon de taille 2500 environ

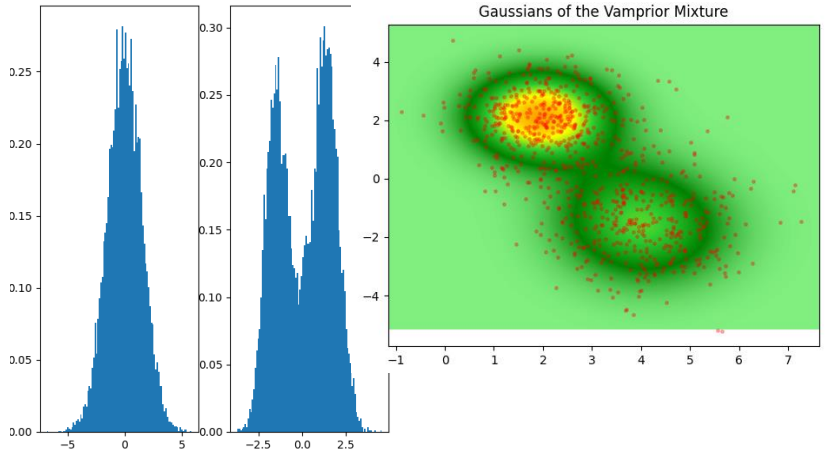
On traite le cas de 4 branches.
Probabilité de défaillance Pf = 0,00093 qui ne dépend pas de la dimension
Pf = P(four_branch(X) > 3,5) avec X ~ N(o, Id)

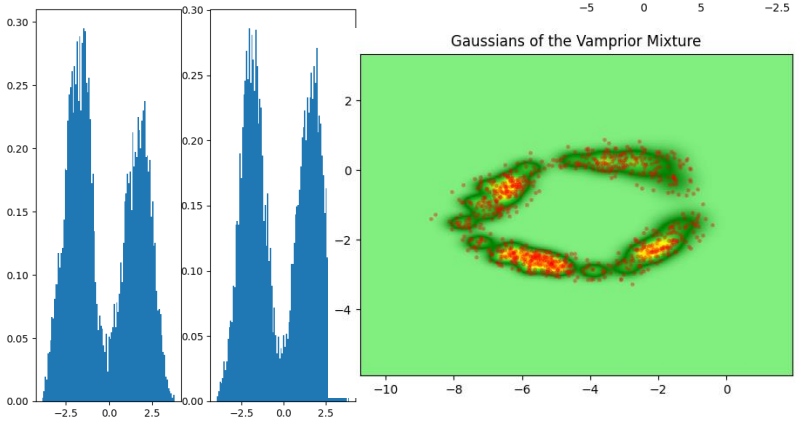


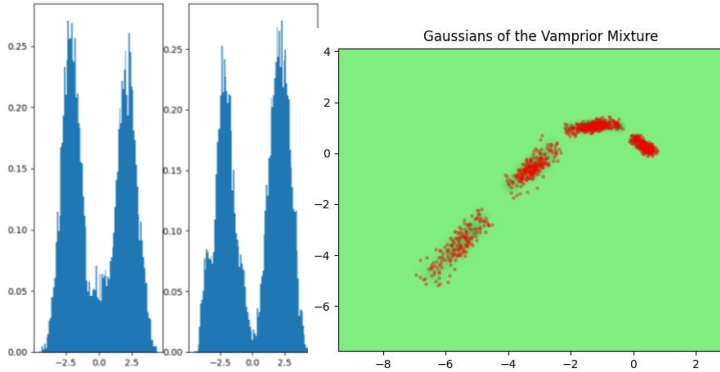
Pf_SS = 0,0009
Sd = 4,19.10^-5

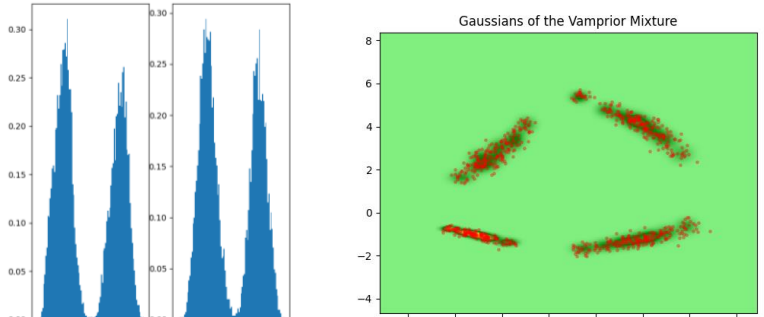


Test du VAE en tant que noyau indépendant
Cas de la dimension 2

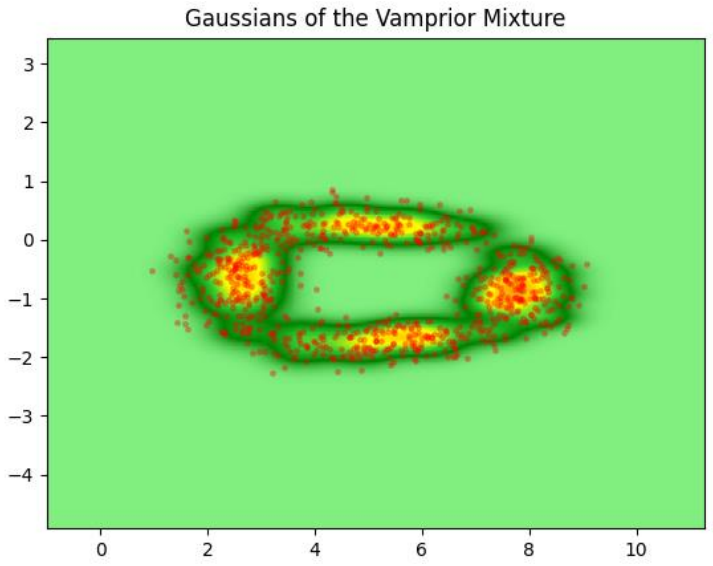


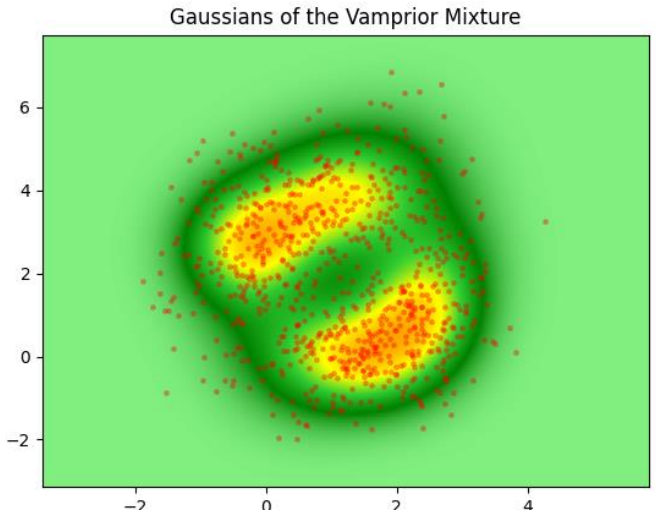


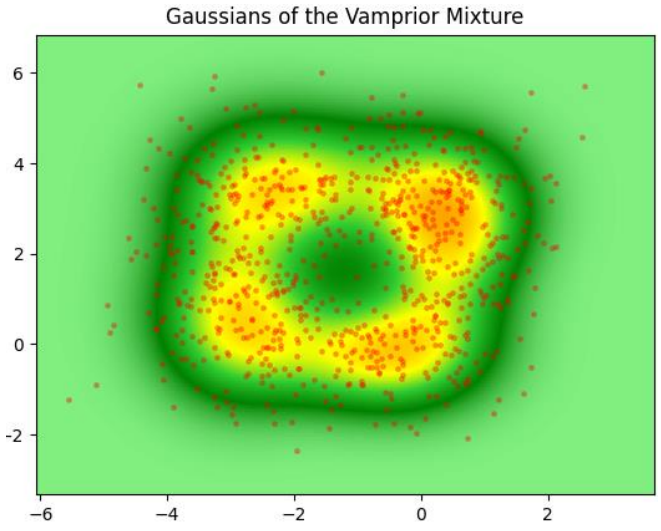


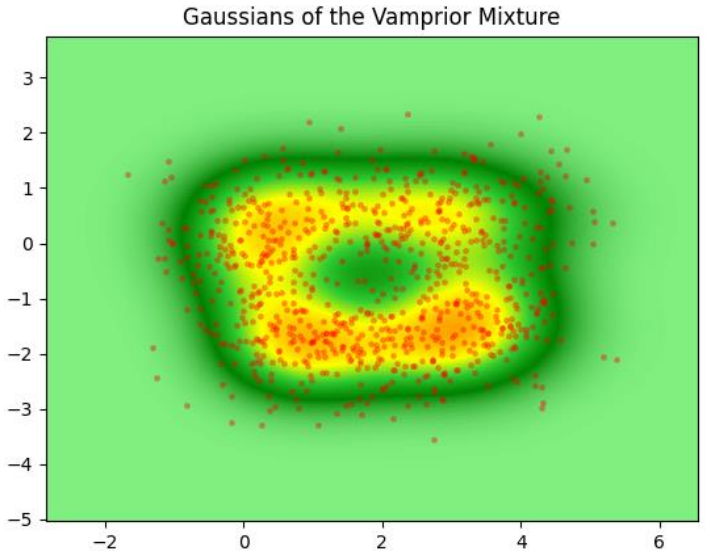


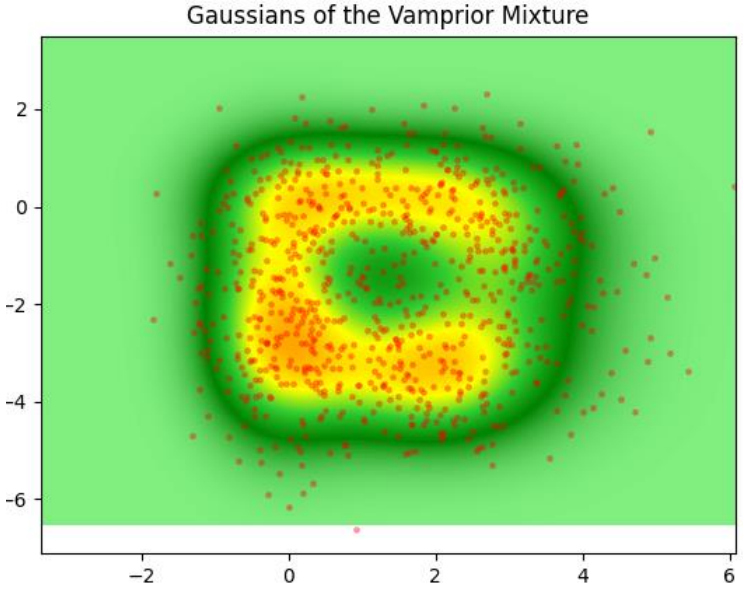
Espace Latent du premier évènement du Subset



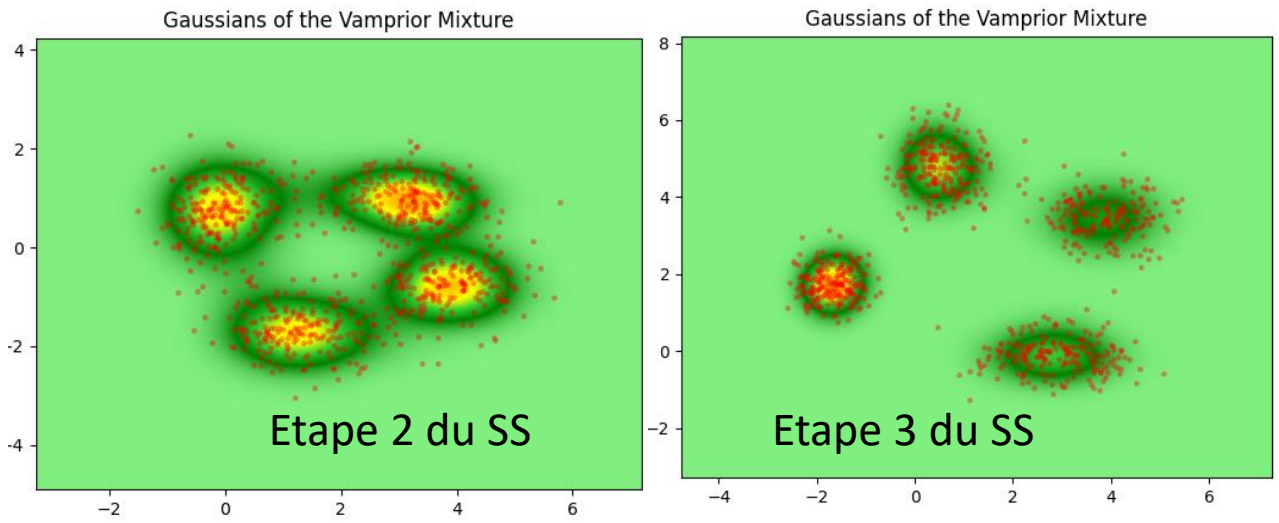








Entrainement sur 2500 échantillons

Méthode	Esperance	Ecart-type	Paramètres	Dimension	Mélange Infini
SS_VAE	Pf = 0,000983	4,098.10^-5	pseudo = 35	10	1000
MMA	Pf = 0,000925	5,32.10^-5	Sd = 0,4	10	--
Vanilla	Pf = 0,000929	5,79.10^-5	Sd =0,4	10	--
CMC	Pf = 0.000914	0,00027	--	10	--

Comportement du VAE :
- Difficile de repérer certaines pour des seuils plus avancées
- Comparaison avec les seuils renvoyés par MMA , toujours une étape de moins. Moins précis ?